

Solução do Problema E226 - Solver HiGHS

Análise Computacional

6 de outubro de 2025

1 Informações do Problema

Informações do Problema:

- Nome: E226
- Número de Variáveis: 282
- Número de Restrições: 223
- Inviabilidade Primal: 0.000e+00
- Inviabilidade Dual: 0.000e+00
- Valor Primal: -1.164e+01
- Valor Dual: -1.875e+01
- Gap: -7.544e-17
- Número de Iterações: 22

2 Variáveis Primais e Custos Reduzidos

Tabela 1: Variáveis primais (x) e custos reduzidos (z) do problema E226

Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
1	1.984e-01	1	0.000e+00
2	9.491e-02	2	0.000e+00
3	4.250e-02	3	0.000e+00
4	0.000e+00	4	0.000e+00
5	2.137e-01	5	0.000e+00
6	0.000e+00	6	0.000e+00
7	0.000e+00	7	5.333e-02
8	5.805e+00	8	0.000e+00
9	7.750e-02	9	0.000e+00
10	0.000e+00	10	0.000e+00
11	1.169e-02	11	0.000e+00
12	0.000e+00	12	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
13	0.000e+00	13	0.000e+00
14	0.000e+00	14	0.000e+00
15	0.000e+00	15	4.066e-02
16	0.000e+00	16	0.000e+00
17	4.870e-02	17	0.000e+00
18	3.987e-01	18	0.000e+00
19	0.000e+00	19	0.000e+00
20	3.943e-01	20	0.000e+00
21	0.000e+00	21	0.000e+00
22	1.061e+01	22	0.000e+00
23	1.585e-01	23	0.000e+00
24	3.966e+00	24	0.000e+00
25	1.964e+00	25	0.000e+00
26	9.600e-01	26	0.000e+00
27	2.908e+00	27	0.000e+00
28	1.786e+00	28	0.000e+00
29	1.019e-01	29	0.000e+00
30	6.045e-01	30	0.000e+00
31	1.265e+00	31	0.000e+00
32	0.000e+00	32	1.590e-02
33	9.721e+00	33	0.000e+00
34	1.500e+00	34	0.000e+00
35	5.251e-02	35	0.000e+00
36	0.000e+00	36	6.028e-01
37	1.346e-01	37	0.000e+00
38	4.207e-01	38	0.000e+00
39	6.665e-01	39	0.000e+00
40	0.000e+00	40	3.428e-01
41	5.095e-02	41	0.000e+00
42	0.000e+00	42	0.000e+00
43	8.414e-02	43	0.000e+00
44	0.000e+00	44	2.108e+01
45	0.000e+00	45	9.000e-04
46	0.000e+00	46	4.642e-01
47	4.290e-02	47	0.000e+00
48	0.000e+00	48	1.000e+00
49	6.303e-02	49	0.000e+00
50	0.000e+00	50	7.471e-01
51	1.815e-01	51	0.000e+00
52	0.000e+00	52	1.466e-02
53	0.000e+00	53	1.017e+00
54	1.938e-01	54	0.000e+00
55	1.664e+00	55	0.000e+00
56	0.000e+00	56	5.052e-01
57	6.344e-01	57	0.000e+00
58	2.874e-01	58	0.000e+00
59	1.664e+00	59	0.000e+00
60	1.140e-01	60	0.000e+00
61	0.000e+00	61	6.518e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
62	3.270e-02	62	0.000e+00
63	0.000e+00	63	6.350e-02
64	1.218e-02	64	0.000e+00
65	0.000e+00	65	1.355e+00
66	2.193e-02	66	0.000e+00
67	9.564e-02	67	0.000e+00
68	4.866e-01	68	0.000e+00
69	1.968e-01	69	0.000e+00
70	9.424e-02	70	0.000e+00
71	5.098e-01	71	0.000e+00
72	1.397e+00	72	0.000e+00
73	1.375e-01	73	0.000e+00
74	0.000e+00	74	1.666e-02
75	3.373e+01	75	0.000e+00
76	1.793e+01	76	0.000e+00
77	3.723e-02	77	0.000e+00
78	1.072e+00	78	0.000e+00
79	3.825e+00	79	0.000e+00
80	1.055e-01	80	0.000e+00
81	4.827e-01	81	0.000e+00
82	1.189e+01	82	0.000e+00
83	2.414e+01	83	0.000e+00
84	7.358e+01	84	0.000e+00
85	9.892e-01	85	0.000e+00
86	0.000e+00	86	2.930e-02
87	1.277e-02	87	0.000e+00
88	0.000e+00	88	1.472e-01
89	0.000e+00	89	1.654e-01
90	0.000e+00	90	5.553e-02
91	1.153e-01	91	0.000e+00
92	0.000e+00	92	6.114e-04
93	6.631e-02	93	0.000e+00
94	1.163e-02	94	0.000e+00
95	1.927e-01	95	0.000e+00
96	8.931e+00	96	0.000e+00
97	1.159e+01	97	0.000e+00
98	3.250e+01	98	0.000e+00
99	6.161e+00	99	0.000e+00
100	1.034e+02	100	0.000e+00
101	4.465e+00	101	0.000e+00
102	1.550e+01	102	0.000e+00
103	6.896e+01	103	0.000e+00
104	0.000e+00	104	5.470e-01
105	0.000e+00	105	1.102e+00
106	0.000e+00	106	6.526e-01
107	0.000e+00	107	4.718e-01
108	0.000e+00	108	2.461e-01
109	2.314e-01	109	0.000e+00
110	0.000e+00	110	1.167e-01

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
111	0.000e+00	111	1.687e-01
112	3.692e+00	112	0.000e+00
113	3.470e-01	113	0.000e+00
114	0.000e+00	114	5.074e-01
115	3.161e-01	115	0.000e+00
116	0.000e+00	116	1.527e+00
117	0.000e+00	117	2.190e-01
118	0.000e+00	118	1.287e-02
119	0.000e+00	119	1.447e-01
120	0.000e+00	120	2.051e-01
121	0.000e+00	121	3.269e-01
122	0.000e+00	122	4.126e-02
123	0.000e+00	123	1.169e-01
124	0.000e+00	124	7.633e-02
125	1.845e-01	125	0.000e+00
126	2.172e-01	126	0.000e+00
127	5.927e-03	127	0.000e+00
128	0.000e+00	128	1.821e-01
129	4.905e-02	129	0.000e+00
130	5.267e-01	130	0.000e+00
131	0.000e+00	131	1.192e+00
132	0.000e+00	132	5.098e-01
133	0.000e+00	133	0.000e+00
134	0.000e+00	134	0.000e+00
135	2.958e-01	135	0.000e+00
136	0.000e+00	136	8.974e-02
137	0.000e+00	137	1.167e-01
138	0.000e+00	138	1.553e-01
139	0.000e+00	139	8.491e-01
140	0.000e+00	140	0.000e+00
141	0.000e+00	141	0.000e+00
142	5.122e-01	142	0.000e+00
143	3.905e-02	143	0.000e+00
144	0.000e+00	144	8.169e-02
145	0.000e+00	145	3.461e-01
146	0.000e+00	146	4.716e-01
147	0.000e+00	147	5.730e-01
148	0.000e+00	148	2.226e-01
149	0.000e+00	149	6.953e-01
150	0.000e+00	150	6.893e-02
151	4.864e-01	151	0.000e+00
152	0.000e+00	152	2.025e-02
153	4.599e-02	153	0.000e+00
154	6.652e-01	154	0.000e+00
155	5.792e-02	155	0.000e+00
156	4.932e-01	156	0.000e+00
157	0.000e+00	157	5.249e-01
158	6.248e-01	158	0.000e+00
159	3.764e-02	159	0.000e+00

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
160	0.000e+00	160	3.235e-01
161	0.000e+00	161	1.182e+00
162	0.000e+00	162	1.041e+00
163	0.000e+00	163	3.376e+00
164	5.921e-01	164	0.000e+00
165	1.269e+00	165	0.000e+00
166	1.293e+00	166	0.000e+00
167	3.993e-01	167	0.000e+00
168	0.000e+00	168	1.228e+00
169	0.000e+00	169	1.161e+00
170	0.000e+00	170	1.128e+00
171	0.000e+00	171	1.124e+00
172	0.000e+00	172	9.314e-02
173	0.000e+00	173	3.078e-02
174	1.332e+00	174	0.000e+00
175	0.000e+00	175	9.765e-02
176	0.000e+00	176	4.025e-02
177	0.000e+00	177	8.130e-03
178	0.000e+00	178	1.078e-01
179	0.000e+00	179	4.078e-02
180	0.000e+00	180	8.620e-03
181	0.000e+00	181	2.026e-02
182	1.585e-01	182	0.000e+00
183	1.105e-02	183	0.000e+00
184	0.000e+00	184	1.749e+00
185	0.000e+00	185	9.483e-01
186	1.123e-02	186	0.000e+00
187	1.516e-01	187	0.000e+00
188	3.247e-01	188	0.000e+00
189	0.000e+00	189	0.000e+00
190	8.894e-01	190	0.000e+00
191	0.000e+00	191	6.897e-02
192	2.923e-01	192	0.000e+00
193	1.214e+00	193	0.000e+00
194	9.692e-02	194	0.000e+00
195	0.000e+00	195	0.000e+00
196	0.000e+00	196	1.064e+00
197	0.000e+00	197	1.670e+00
198	0.000e+00	198	1.783e+00
199	0.000e+00	199	1.747e+00
200	0.000e+00	200	1.246e+00
201	1.890e-01	201	0.000e+00
202	0.000e+00	202	4.517e-01
203	0.000e+00	203	5.416e-01
204	0.000e+00	204	7.351e-02
205	4.860e-02	205	0.000e+00
206	0.000e+00	206	9.287e-02
207	0.000e+00	207	8.810e-02
208	0.000e+00	208	2.549e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
209	0.000e+00	209	1.923e-02
210	9.486e-02	210	0.000e+00
211	2.518e-01	211	0.000e+00
212	2.666e-01	212	0.000e+00
213	0.000e+00	213	1.635e-01
214	0.000e+00	214	3.269e-01
215	0.000e+00	215	3.853e-01
216	1.090e-01	216	0.000e+00
217	0.000e+00	217	5.867e-02
218	0.000e+00	218	2.091e-02
219	0.000e+00	219	4.129e-02
220	1.985e-01	220	0.000e+00
221	0.000e+00	221	5.958e-02
222	0.000e+00	222	6.238e-02
223	1.733e-02	223	0.000e+00
224	0.000e+00	224	1.434e-02
225	0.000e+00	225	4.348e-02
226	0.000e+00	226	4.129e-02
227	0.000e+00	227	2.527e-04
228	9.006e-01	228	0.000e+00
229	0.000e+00	229	9.781e-03
230	0.000e+00	230	1.650e-01
231	0.000e+00	231	2.215e-03
232	6.900e-01	232	0.000e+00
233	0.000e+00	233	5.712e-03
234	0.000e+00	234	1.646e+00
235	1.061e+00	235	0.000e+00
236	0.000e+00	236	5.781e-01
237	0.000e+00	237	1.521e-01
238	0.000e+00	238	1.886e-02
239	0.000e+00	239	1.981e-02
240	0.000e+00	240	7.857e-01
241	0.000e+00	241	4.566e-03
242	1.254e+00	242	0.000e+00
243	0.000e+00	243	1.466e-02
244	0.000e+00	244	1.781e-02
245	0.000e+00	245	1.371e-02
246	0.000e+00	246	4.334e-02
247	0.000e+00	247	7.883e-02
248	0.000e+00	248	2.783e-02
249	0.000e+00	249	1.370e-02
250	0.000e+00	250	4.667e-01
251	3.033e+00	251	0.000e+00
252	1.218e+00	252	0.000e+00
253	5.095e-01	253	0.000e+00
254	0.000e+00	254	1.637e-01
255	0.000e+00	255	5.071e-04
256	2.736e+00	256	0.000e+00
257	0.000e+00	257	3.989e-02

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Coordenada x	Valor x (Primal)	Coordenada z	Valor z (Custo Reduzido)
258	0.000e+00	258	9.550e-01
259	1.587e+00	259	0.000e+00
260	5.290e-01	260	0.000e+00
261	0.000e+00	261	1.072e-01
262	2.264e+00	262	0.000e+00
263	0.000e+00	263	1.495e-01
264	0.000e+00	264	1.282e-01
265	0.000e+00	265	4.630e-02
266	0.000e+00	266	3.217e-02
267	0.000e+00	267	1.330e-01
268	6.295e-01	268	0.000e+00
269	0.000e+00	269	9.234e-03
270	0.000e+00	270	1.146e-02
271	4.284e+00	271	0.000e+00
272	0.000e+00	272	1.221e-02
273	0.000e+00	273	5.119e-02
274	0.000e+00	274	5.543e-02
275	4.563e-01	275	0.000e+00
276	0.000e+00	276	8.284e-03
277	0.000e+00	277	4.822e-03
278	4.410e-01	278	0.000e+00
279	0.000e+00	279	1.187e-02
280	0.000e+00	280	8.956e-03
281	0.000e+00	281	2.262e+00
282	0.000e+00	282	0.000e+00

3 Variáveis Duais (Multiplicadores de Lagrange)

Tabela 2: Variáveis duais (y) do problema E226

Coordenada y	Valor y (Dual)
1	0.000e+00
2	7.543e-02
3	0.000e+00
4	-1.877e-03
5	6.035e-02
6	7.396e-02
7	0.000e+00
8	0.000e+00
9	-2.401e-02
10	-3.015e-02
11	0.000e+00
12	0.000e+00
13	0.000e+00
14	-1.266e+00
15	0.000e+00

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Coordenada y	Valor y (Dual)
16	-3.643e+00
17	1.078e+00
18	-4.188e+00
19	1.078e+00
20	0.000e+00
21	0.000e+00
22	-4.942e+00
23	1.078e+00
24	0.000e+00
25	-3.521e+00
26	-1.022e+00
27	-1.800e-01
28	-3.350e+00
29	-1.535e-01
30	-1.885e-01
31	0.000e+00
32	-2.617e+00
33	-1.767e-01
34	0.000e+00
35	-4.044e+00
36	-3.572e+00
37	-4.062e+00
38	-7.214e-01
39	-3.322e-01
40	-8.120e+00
41	0.000e+00
42	-8.700e-03
43	-4.353e+00
44	0.000e+00
45	0.000e+00
46	-1.078e+00
47	0.000e+00
48	0.000e+00
49	0.000e+00
50	-5.509e-02
51	0.000e+00
52	-2.116e-02
53	-6.995e-01
54	0.000e+00
55	0.000e+00
56	0.000e+00
57	0.000e+00
58	-9.978e-01
59	-3.350e+00
60	-3.347e+00
61	0.000e+00
62	-3.350e+00
63	-6.700e-01
64	-4.990e-01

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Coordenada y	Valor y (Dual)
65	-3.321e+00
66	0.000e+00
67	-3.260e+00
68	-2.999e+00
69	-3.250e+00
70	-3.240e+00
71	-3.197e+00
72	-3.193e+00
73	-3.271e+00
74	-3.200e+00
75	-3.262e+00
76	-2.527e-02
77	-2.843e+00
78	-4.527e+00
79	-3.121e+00
80	-5.800e-02
81	0.000e+00
82	0.000e+00
83	0.000e+00
84	-2.843e+00
85	2.285e-01
86	-2.527e-02
87	-1.506e+00
88	-1.646e+00
89	-1.208e+00
90	-2.294e-01
91	0.000e+00
92	0.000e+00
93	-1.514e+00
94	0.000e+00
95	-6.725e-01
96	-1.475e+00
97	2.200e-02
98	2.750e-03
99	-2.294e+00
100	-1.908e+00
101	-1.428e+00
102	0.000e+00
103	-1.620e+00
104	4.788e-01
105	0.000e+00
106	-9.902e-02
107	2.200e-03
108	-1.482e-02
109	-7.800e-03
110	0.000e+00
111	0.000e+00
112	0.000e+00
113	0.000e+00

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Coordenada y	Valor y (Dual)
114	0.000e+00
115	0.000e+00
116	-2.149e+00
117	-2.262e+00
118	-2.294e+00
119	-9.105e-02
120	-1.514e+00
121	0.000e+00
122	-1.284e+00
123	0.000e+00
124	0.000e+00
125	-1.172e+00
126	0.000e+00
127	0.000e+00
128	-4.252e-03
129	0.000e+00
130	-2.386e+00
131	-2.495e+00
132	-3.495e+00
133	0.000e+00
134	0.000e+00
135	0.000e+00
136	0.000e+00
137	0.000e+00
138	-4.749e-03
139	-1.016e-01
140	-3.954e-03
141	0.000e+00
142	-2.359e+00
143	-2.989e+00
144	-3.135e-03
145	0.000e+00
146	0.000e+00
147	0.000e+00
148	0.000e+00
149	0.000e+00
150	-6.879e-02
151	-2.854e-03
152	0.000e+00
153	-1.950e+00
154	-2.500e+00
155	0.000e+00
156	0.000e+00
157	0.000e+00
158	0.000e+00
159	0.000e+00
160	0.000e+00
161	-1.317e-02
162	-5.501e-03

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Coordenada y	Valor y (Dual)
163	0.000e+00
164	-6.130e-02
165	0.000e+00
166	0.000e+00
167	0.000e+00
168	0.000e+00
169	0.000e+00
170	0.000e+00
171	-2.194e-01
172	-1.886e+00
173	-1.878e+00
174	-1.884e+00
175	-1.901e+00
176	-2.826e+00
177	-2.976e+00
178	-3.097e+00
179	-3.229e+00
180	-2.294e+00
181	-1.164e-03
182	-1.164e-03
183	0.000e+00
184	-2.294e+00
185	0.000e+00
186	-2.526e+00
187	0.000e+00
188	-1.032e+01
189	-7.758e-02
190	-1.012e+01
191	-2.912e+01
192	-1.939e-02
193	-1.514e+00
194	-2.526e+00
195	-1.845e+00
196	1.834e-02
197	-1.521e-03
198	-3.393e-03
199	0.000e+00
200	-2.274e+00
201	-2.414e+00
202	-2.366e+00
203	-3.206e+00
204	0.000e+00
205	0.000e+00
206	0.000e+00
207	-3.635e-02
208	0.000e+00
209	1.203e-02
210	-1.203e-02
211	0.000e+00
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Coordenada y	Valor y (Dual)
212	0.000e+00
213	0.000e+00
214	-4.074e-01
215	0.000e+00
216	-5.369e-01
217	0.000e+00
218	0.000e+00
219	-9.815e-02
220	-1.378e+00
221	-3.521e+00
222	-3.014e+00
223	-2.242e+00